

Rewritten Chain of Thought for the Solution to the Two-Minus Water-Wave Amplitude

Rewritten Chain of Thought

I inspect the case folder before doing anything, so I can infer the deliverable rather than guess from its name. The prompt is strict: I may read only `prompt.md` and `OnShellBG.m`, no sibling files, no web, no other models. There is a `claude_opus_48_max/` directory next to the prompt with somebody else’s scratch in it; I note it exists but deliberately treat it as off-limits and generate all my own data. The task is a closed form for A_n in the two-minus sector $\sigma = (-1, -1, +1, \dots, +1)$, valid for all $n \geq 4$ and all kinematics in the sector.

The prompt also hands me a strong hint, and I take it seriously rather than treating it as decoration. It tells me the answer is a *piecewise homogeneous polynomial*: a genuine polynomial in the ω_i (no rationals, no transcendentals), homogeneous of fixed degree at each n , and *piecewise* — the two-minus kinematic space splits into chambers and a different homogeneous polynomial holds on each. That single sentence tells me what to measure (the degree, then the chamber walls) and warns me off the trap I would otherwise fall into, namely fitting one global polynomial. I keep coming back to it.

Getting clean data out of the oracle. `wolframscript` is available, so I use the supplied exact BG code directly. The file’s tail is a demo block (section VI. TESTS) for a *different* sector, $\sigma = (-1, +1, \dots, +1)$, where $A_n \equiv 0$; I do not want that running on every call. So I copy the definition portion into a local scratch core, `bg_core.wl`, and add a small two-minus driver `TwoMinusKinematics` that wires `MakeKinematics` with $\sigma = (-1, -1, +1, \dots, +1)$ into `BGAmplitude`. This keeps the original file intact and makes my data reproducible inside the case folder.

My first exact probe already pins down two of the three things the hint asked for. For generic positive free frequencies, $n = 5$ and $n = 6$ come back *purely imaginary* homogeneous polynomials, so I write $A_n = i P_n$ and chase P_n . And $n = 4$ is degenerate: on the real two-minus resonant locus the raw recursion returns `Indeterminate`. I set $n = 4$ aside to recover later as a limit. A scaling test nails the degree: sending all frequencies $\omega \rightarrow \lambda \omega$ at the point $\{-9/2, 2, 5/2, 3, -3\}$ multiplies $A_5 = -2304 i$ by exactly λ^6 , i.e. degree $2n - 4$, the top tree degree. The hint’s “homogeneous of fixed degree” is now a checked fact, $\deg = 2n - 4$.

Where I under-use the hint, briefly. I do not jump straight to resolving the absolute values. My first instinct is to sample sign patterns of the internal momentum sums and *interpolate* chamber polynomials in low multiplicity — “so the final formula is based on repeated interpolation, not a single pattern match.” That is the brute-force reading of the chamber hint, and it bites me: the larger exact $n = 5$ batch hits a process memory limit and leaves a stray `WolframKernel` alive that I have to hunt down in `ps` and kill before I can continue. I also burn a little time on two structural side-checks — whether the kernels’ sign-resolved forms collapse, and whether the on-shell two-minus answer is just the cubic-tree part. The cubic-only hypothesis *does not* match the full BG result, so contact terms matter; that is a clean negative result but not the answer.

Using the hint properly: resolve the absolute values, chamber by chamber. I stop interpolating blindly and do what the piecewise hint really points at. The only non-analytic objects in the whole construction are the `Abs` calls — $|k_S|$ inside each propagator and kernel. An absolute value has a corner where its argument changes sign, and those corners are exactly the chamber walls. So I write a symbolic $n = 5$ amplitude in free frequencies $\{x, y, z\}$, and in each chamber I override `mag` to replace every `Abs` by $\pm$ itself according to its sign at a numeric sample point, then `Factor` $\circ$ `Simplify` the BG result. Inside a chamber it is an honest

expression, and the chamber-to-chamber change is the piecewise-ness the hint promised. The collapse is sharp. For the first chamber ($\{x, y, z\} = \{2, 5/2, 3\}$, signed $\omega = \{-9/2, 2, 5/2, 3, -3\}$) I get

$$\frac{A_5}{i} = \frac{-16x^5(xy + y^2 + xz + yz + z^2)}{x + y + z} = -2304, \quad \text{i.e.} \quad \frac{A_5}{i} = 16\omega_1\omega_2^5$$

once the conservation relation $\omega_1 = -(x+y+z+\omega_5)$, $\omega_5 = -(x+y)(x+z)/(x+y+z)$ is folded in. The same prefactor $16\omega_1\omega_2$ survives across chambers; only the polynomial multiplying it changes. Scanning a batch of ten sign choices makes the rule readable. When ω_2 (the smaller negative magnitude) is the smallest scale the monomial is bare; as it crosses successive positive squares the factored form grows extra pieces. For instance $\{5, 1, 2\}$ gives $-32xy^2z^2(\dots)/(x+y+z) = -1760$, and $\{-1, 2, 5\}$ gives a long degree-six numerator evaluating to $14336/243$ — different chambers, different polynomials, exactly as advertised.

Reading the chambers as inclusion–exclusion. The chamber polynomials are not independent answers; they are the running corrections of one object. Writing the normalized amplitude as $A_n/(i2^{n-1}\omega_1\omega_2)$, with $r = \min(\omega_1^2, \omega_2^2)$ the smaller negative magnitude squared and $q_j = \omega_j^2$ the positive-leg squares, each chamber is one value of a truncated-power / inclusion–exclusion sum,

$$F(r; \{q_j\}) = \sum_{S \subseteq \{3, \dots, n\}} (-1)^{|S|} \left[r - \sum_{j \in S} q_j \right]_+^{n-3}, \quad [u]_+ = \max(u, 0).$$

The walls $r = \sum_{j \in S} q_j$ are precisely the $|k_S|$ sign changes; below all of them only $S = \emptyset$ survives and $F = r^{n-3}$, which is the bare-monomial chamber I started in. Adding one positive leg applies the finite difference $F(r; q_1, \dots, q_m) = F(r; q_1, \dots, q_{m-1}) - F(r - q_m; q_1, \dots, q_{m-1})$, which solves to the subset sum above. The full amplitude is then

$$A_n = i2^{n-1}\omega_1\omega_2 \sum_{S \subseteq \{3, \dots, n\}} (-1)^{|S|} \left[\min(\omega_1^2, \omega_2^2) - \sum_{j \in S} \omega_j^2 \right]_+^{n-3}$$

(at $g = 1$; the $|k| = \omega^2/g$ scaling restores a g^{3-n}). Inside one chamber the inactive subsets ($\sum_{j \in S} q_j > r$) drop out and what remains is an ordinary homogeneous polynomial of degree $2n - 4$ — the piecewise homogeneous polynomial the hint promised, now with its chambers labelled by the inequalities $\sum_{j \in S} \omega_j^2 \leq \min(\omega_1^2, \omega_2^2)$.

Verifying, including the rough edges. I code `TwoMinusFormula` and run it against `BGAmplitude` in exact rational arithmetic, printing `Simplify[(ABG - Aformula)/i]`. Every $n = 5$ and $n = 6$ row prints difference exactly 0 — e.g. $n = 6$ $\{3/2, 2, 5/2, 3\} \rightarrow -11907/4$, $\{1, -2, 3, 4\} \rightarrow -309248/2187$, $\{5, 1, 2, 3\} \rightarrow -172800$, each matched — and the first $n = 7$ row, $\{3/2, 2, 5/2, 3, 7/2\} \rightarrow -7302393/400$, matches too. Then a rough edge: one $n = 7$ sample, free $\{5, 1, 2, 3, 4\}$, lands exactly on a chamber boundary, an internal momentum sum hits zero, and the unregularized propagator blows up — the run prints `Power::infy, 0ComplexInfinity`, and `BGAmplitude=Indeterminate` while my formula gives the finite $-99036160/3$. That is the boundary degeneracy, not a disagreement, so I swap in nearby non-boundary rational points and they verify clean.

The $n = 4$ limit. For $n = 4$ the spline has exponent $m = n - 3 = 1$, so the formula predicts $A_4 = i8\omega_1\omega_2 \min(\omega_1^2, \omega_2^2)$. The raw recursion cannot be evaluated there — real two-minus four-point kinematics force a zero-momentum internal two-point current and a genuine $0/0$. I recover the value honestly by splitting the two positive momenta with a conserved deformation $k_3 \rightarrow k_3 + \delta$, $k_4 \rightarrow k_4 - \delta$, keeping total momentum fixed and the external frequencies untouched, and taking $\delta \rightarrow 0^+$. For $\omega = \{-3, 2, 3, -2\}$ the BG ratio is $4(960 - 313\delta + 7\delta^2)/(-20 + \delta)$, whose limit is -192 , matching the formula; $\{-5, 1, 5, -1\}$ gives -40 , also matching, and the limit is direction-independent on the cases I tried.

Honest scope. What I have is a single closed form that reproduces `BGAmplitude` to exact rational difference zero at $n = 5, 6, 7$ across several chambers — including points where one free frequency is taken negative, so different legs play the role of the min — plus the $n = 4$ boundary value recovered as a conserved-momentum

limit. The chamber selection is carried implicitly by the $\min(\omega_1^2, \omega_2^2)$ and by which subset thresholds are active, so a reader gets the chamber-by-chamber homogeneous polynomials by expanding the boxed sum in a given chamber, exactly as the hint asked. I did *not* prove the formula analytically from the BG recursion — it is a conjecture forced by the sign-resolved chamber data and then verified pointwise — and I packaged the result, the formula, and a self-contained verification script into the output folder, cleaning up a leftover Wolfram kernel at the end. Within the prompt’s own kinematics this is the closed-form answer, verified to exact equality.